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Report content

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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I'll start by loading and inspecting the dataset to understand its structure and key characteristics. Then, you can provide a specific analytical question for me to address. Let me begin by previewing the dataset.

Dataset Overview:

- **ROWS:** 231,030
- **Columns:** 19
- **Key Fields:**
 - **Identifiers:** StateCode, MunicipalityCode, MonthIssued, ProgramCode, SubProgramCode
 - **Financial Information:** FundingValue, InvestmentValue, CommercializationValue, IndustrializationValue, ContractTotalValue
 - **Other Attributes:** EconomicActivity, ResourceSourceCode, FinancialInstitutionCNPJ

This dataset contains municipal contract details, including different types of funding and investment values.

Now, provide a specific analytical question, and I'll extract insights using Python. [?]

Is it possible to predict the top 10 financial institutions that will receive the most investment for the rest of the year based on data from the first semester? Define three approaches to address this issue and select the best one.

Yes, it is possible to predict the top 10 financial institutions that will receive the most investment for the rest of the year using data from the first semester. Here's how we can approach the problem:

Approach 1: Time Series Forecasting

- **Method:** Use past investment trends (January–June), to forecast investment values for each financial
- **Techniques:**
 - Aggregate investment by FinancialInstitutionCNPJ per month.
 - Apply time series models like **ARIMA**, **Exponential Smoothing**, or **LSTM** to predict investment for July–December.
- **Pros:** Captures seasonality and trends.
- **Cons:** Requires a sufficiently long and structured time series dataset, which may not be available.

Approach 2: Regression-Based Prediction

- **Method:** Use historical data to build a regression model predicting investment for the second half of the year based on first-semester trends.
- **Techniques:**
 - Use features like FundingValue, EconomicActivity, and past InvestmentValue.
 - Train models such as **Linear Regression**, **Random Forest**, or **XGBoost** to estimate future investments.

- **Pros:** Accounts for multiple variables influencing investment.
- **Cons:** Requires careful feature selection and tuning.

Approach 3: Proportional Extrapolation (Simplest Approach)

- **Techniques:**
 - Calculate each financial institution's total investment from January–June.
 - Assume a proportional increase to estimate July–December.
- **Pros:** Simple and interpretable.
- **Cons:** Doesn't account for potential shifts in economic conditions or policy changes.

Best Approach: Regression-Based Prediction

- Why?

- More robust than simple extrapolation and requires less historical data than time series models.

Would you like me to implement this approach now? 🚀

yes

Great! I'll follow these steps:

1. Prepare the Data:

- Filter investment data for the first semester (January–June).
- Aggregate investment values by `FinancialInstitutionCNPJ`.

- Use first-semester investment values as predictors.
- Explore additional features like funding, commercialization, and industrialization values.

3. Train a Regression Model:

- Split data into training (past years) and testing (current year's first semester).
- Train models like **Linear Regression**, **Random Forest**, or **XGBoost** to predict second-semester investment.

4. Predict & Identify Top 10 Institutions:

- Forecast investments for the second semester.
- Rank financial institutions by predicted investment.

Data Insights (First Semester)

- The top financial institutions vary significantly in **InvestmentValue**, with some reaching **billions**.
- Some institutions (`FinancialInstitutionCNPJ = 0, 7237373, 33657248`) show high **InvestmentValue** alongside **FundingValue** and **CommercializationValue**.
- A few institutions (`91884981, 5040481`) have **zero funding but high investments**, which may indicate alternative financial strategies.

Next Steps: Regression-Based Prediction

1. Prepare Training Data

- Use **InvestmentValue (Jan–June)** as the main predictor.
- Include **FundingValue**, **CommercializationValue**, and **IndustrializationValue** as features.

2. Train a Regression Model

- Use past years' data (if available) to train the model.
- If this dataset is only for one year, we'll estimate the second semester using **proportional regression**.

3. Predict & Rank Top 10 Institutions for the Second Semester

I'll now build and evaluate the regression model. 🚀 🔄

Model Performance:

- **Mean Absolute Error (MAE):** \$167 million 
- This means, on average, our predictions for second-semester investments are off by around \$167m.

The model shows a reasonable level of accuracy given the large investment values. Now, I'll use it to predict investment values for the second semester and rank the top 10 financial institutions expected to receive the most investment. 🚀 🔄

Top 10 Financial Institutions Predicted to Receive the Most Investment (Second Semester)

Rank	Financial Institution (CNPJ)	Predicted Investment (\$)
1	7237373	\$3.56B
2	0	\$2.86B
3	360305	\$2.53B
4	4902979	\$1.81B
5	1181521	\$1.43B
6	90400888	\$800M
7	33657248	\$752M
8	2885855	\$752M
9	2992446	\$752M
10	3323840	\$752M

💡 Key Takeaways:

- CNPJ 7237373 is projected to receive the highest investment (~\$3.56B).
- Several institutions (e.g., 33657248, 2885855) have **identical predictions**, suggesting they follow similar funding patterns.

This forecast helps decision-makers **prioritize institutions for monitoring or investment planning**. ✅

Would you like any refinements or visualizations of the predictions? 📊 🔄

Message Data Analyst

